

Using a Reduced-order Dynamical Model to Determine the Effect of Posture on Knee Loading During Jump Landing

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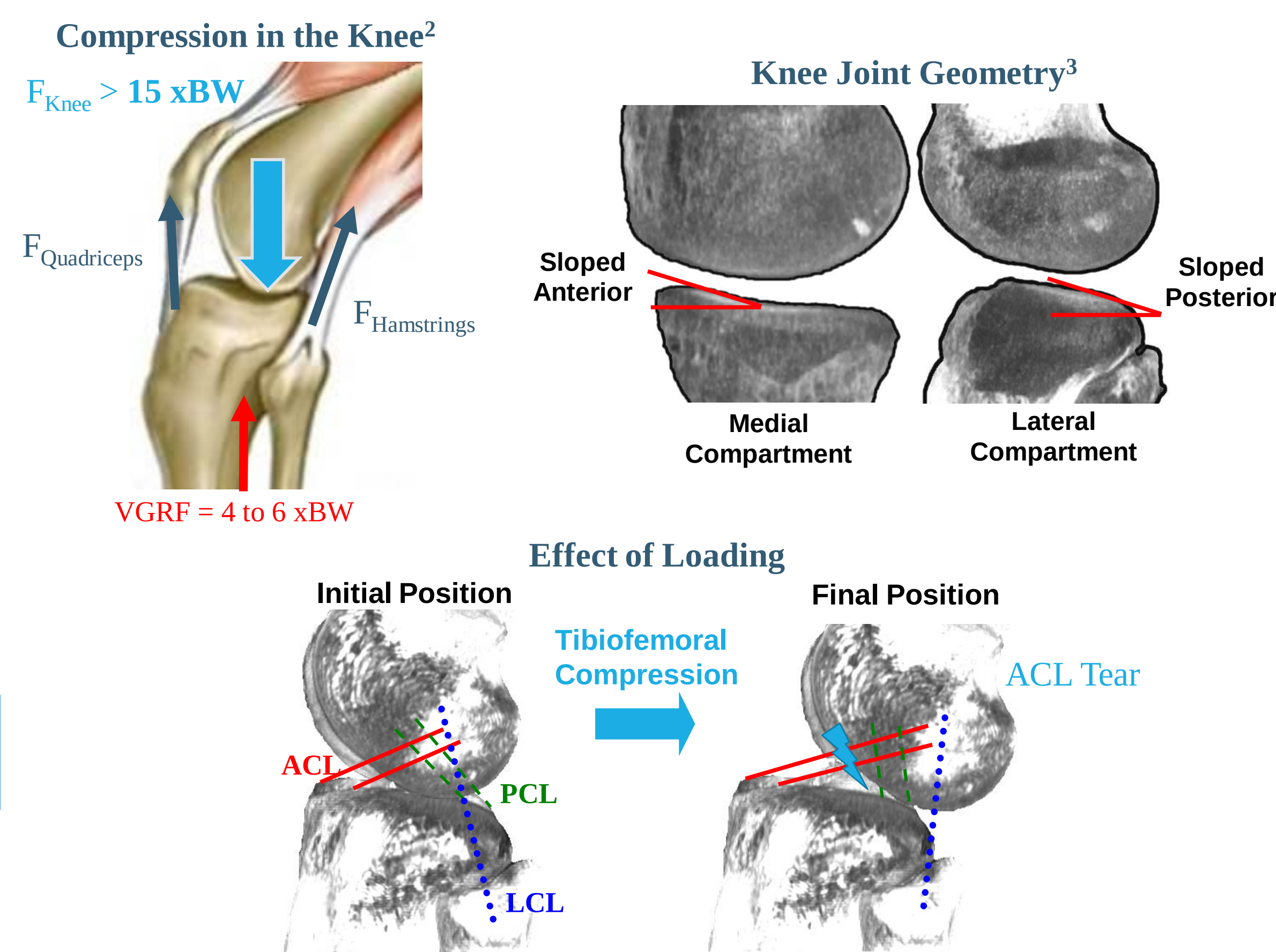
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Abstract P1653
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Introduction

- Excessive knee loading during jump landing is an important injury mechanism, particularly for the anterior cruciate ligament.¹



- Sagittal plane postures of the lower extremities and torso can modulate the force vector through the knee joint.^{4,5}
- A dynamical modelling method has been advocated where the simplest model (least number of variables and parameters) that exhibit specific behavior is used to understand control strategies.⁶

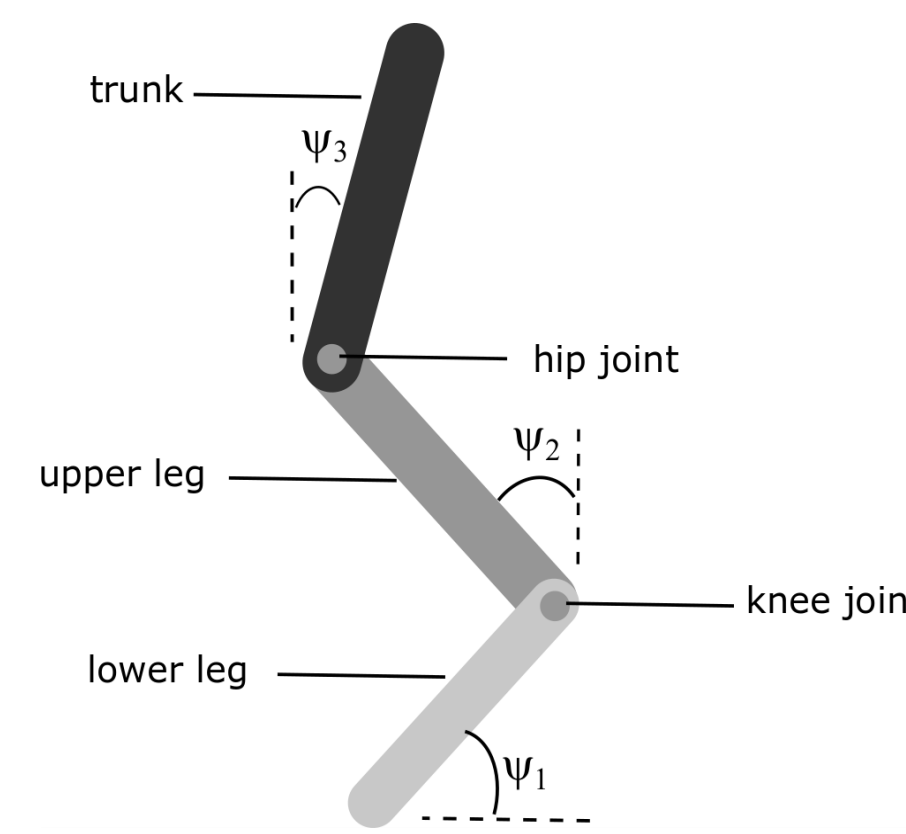
Project Goals

- Develop a planar, reduced-order, multi-body, dynamical model of the human lower extremity and torso.**
- Find postures that minimize knee loading during landing from a jump.**

Methods

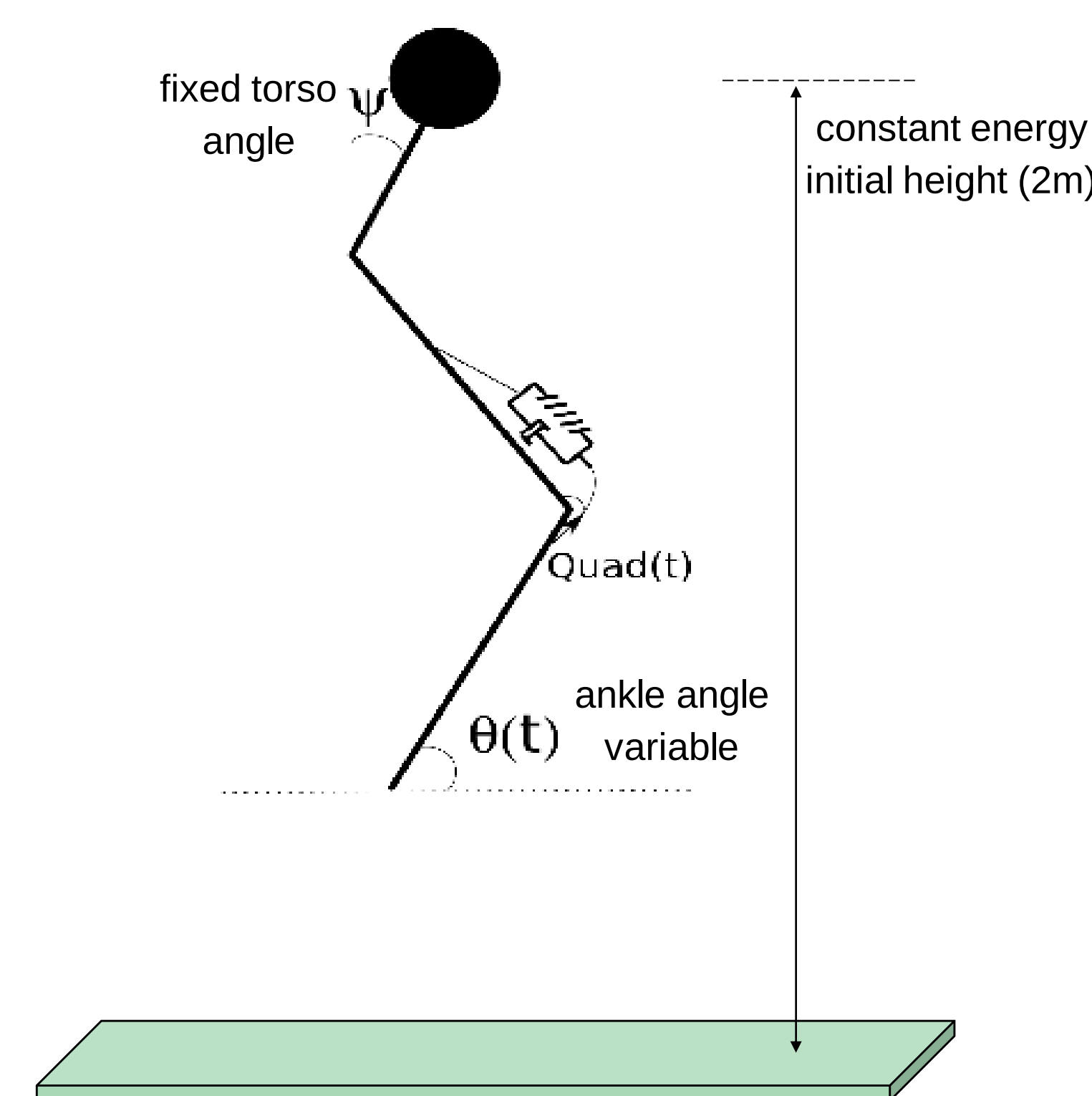
- The model included a mass and three segments (torso, upper leg, lower leg).

Model Parameters
Mass and lengths were based on previous human-scale characteristics from bipedal gait simulations.⁷



- Dynamical Simulation Details**

- Quadriceps activation applied with a passive spring/dashpot force system Quad(t) at the knee.
- The torso angle Ψ was fixed relative to the vertical axis during the simulation, while the ankle angle $\theta(t)$ was initially set at $t=0$, but varied by the dynamical control during the landing.
- All simulations had the same initial center of mass height set to 2 m (constant energy).

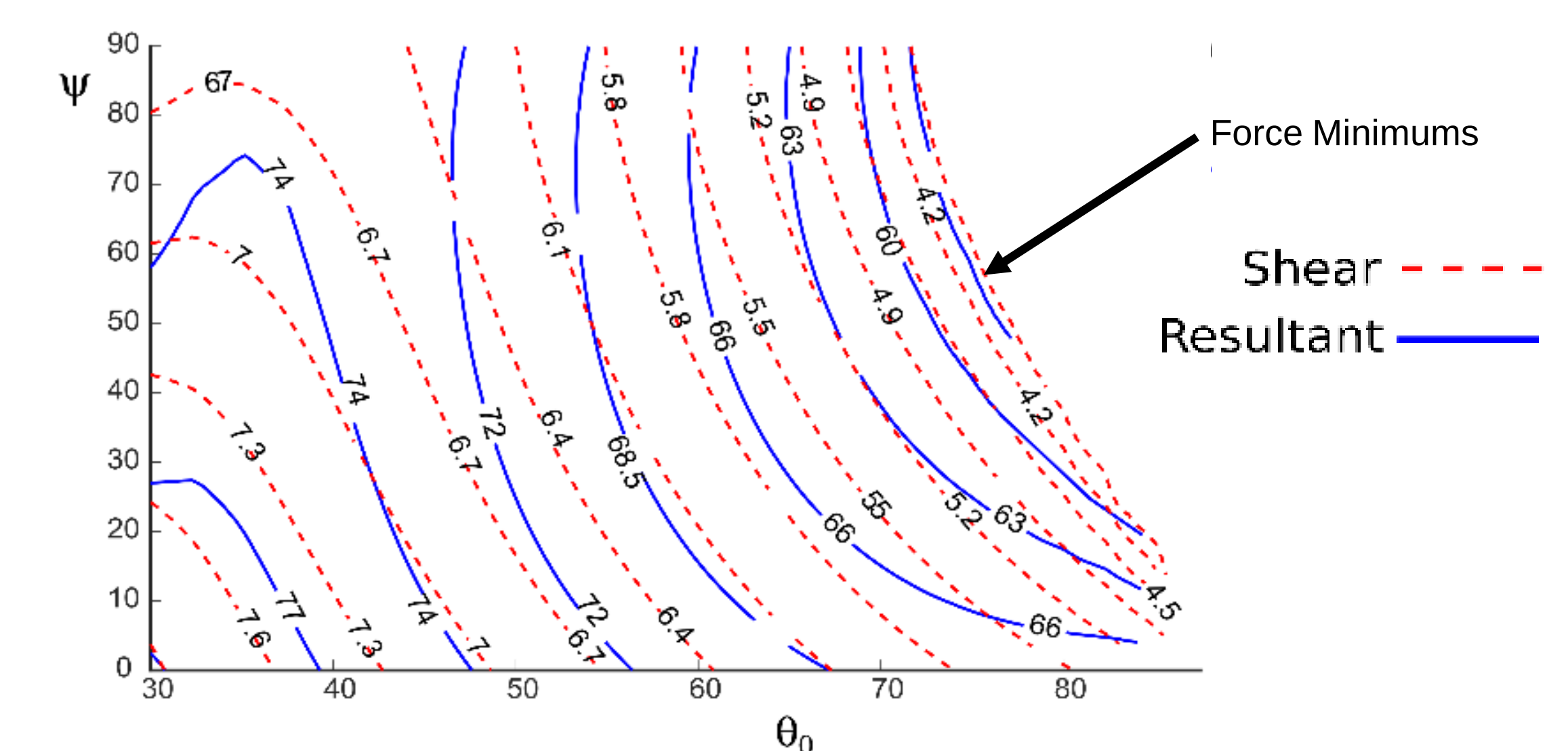


- The relative muscle stiffness and damping ratio was changed and the effects were compared for various postures.
- The maximum shear and whole resultant forces on the knee were reported during the simulations.

Results

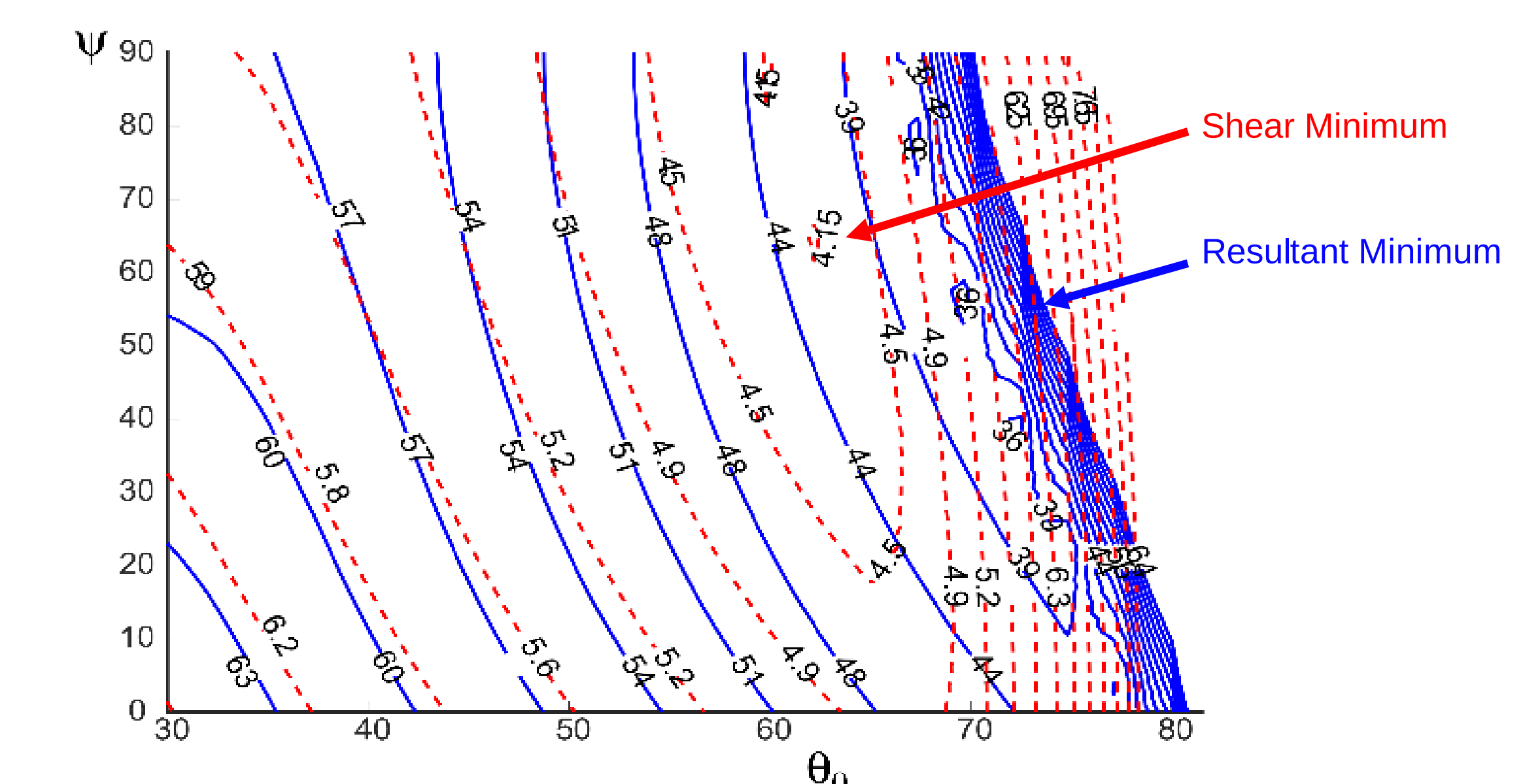
- Undamped Model**

- Predicts a minimum force in the knee (BW) of $F_{\text{Shear}}=4.0 \times \text{BW}$ and $F_{\text{Resultant}}=58$ at torso and ankle angles each approximately 70-85°.



- Damped Model**

- Predicts different optimal solutions of approximately $F_{\text{Shear}}=4.2$ and $F_{\text{Resultant}}=36$ for shear force at $\Psi=70^\circ$ and $\theta_0=65^\circ$ and resultant force at $\Psi=60^\circ$ and $\theta_0=70^\circ$.



Discussion

- Two strategies control landing forces.
 - The posture can be positioned during the flight phase in preparation for initial ground contact.
 - The stiffness and damping could be varied through neuromuscular control during the stance phase.
- Optimal postures were highlighted but other postures that occur during landing could lead to injury.
- A narrower range of optimal postures and different shear or resultant minimums occur with damping.